Review Session:

Venture Finance

Penn Engineering

EAS5450

Exam Format - All Sections

Tuesday, May 6th, 12-2PM

- **Section 1: Short Answer (80%)**
  - Answer 16 of 20 questions requiring brief (1-4) sentence response and/or calculations
  - Bullets OK if question asks for list
- **Section 2: VC Evaluation (20%)**
  - Choose from two term sheet offers
  - Answer all 8 of 8 questions
  - Quantitative analysis
  - Qualitative analysis

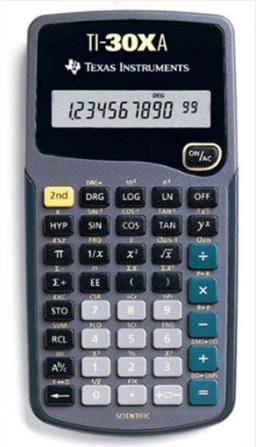
Respondus LockDown Browser Required

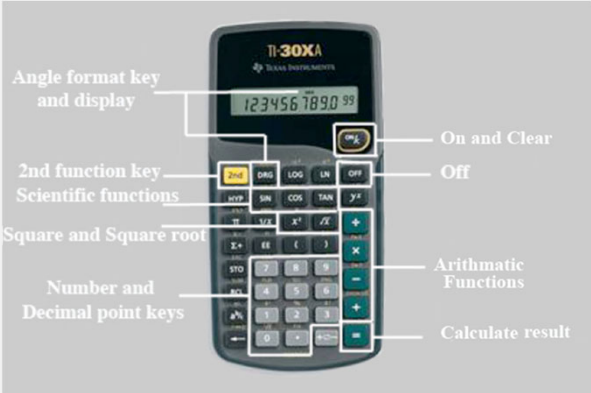
- Make sure Respondus LockDown Browser is installed, configured, and tested on your laptop.
- Make sure your laptop has at least 2 hours of charge:
  - Electric outlets may not be available.

EAS5450 Final Exam: Preparing …

- The exam is closed book except you may refer to two 2-sided, 8½ x 11sheets (4 pages) of personal notes: hand-written, cut-and-pasted, or printed.
- Review key take-aways from:
  - Case studies
  - Assigned readings
  - Venture Capital Method
  - Lecture slides and notes
  - Guest speaker advice and lessons-learned

A calculator will be provided for use during the exam (model TI-30XA). Personal calculators may not be used. You may try the calculator in advance to become familiar with it (recommended). See Ashley in 304 Towne.



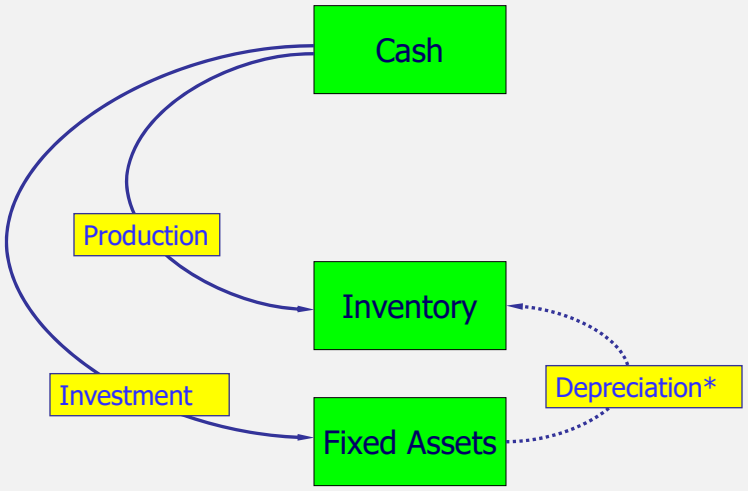


Venture Finance

Review Session Agenda

- Cash flow cycle
- Bootstrapping
- Venture capital
- Exit strategies: IPOs & mergers/acquisitions
- Financial ratios & cash conversion cycle
- Pricing
- Venture Capital Problem Set
- Pitching to investors
- Term sheets

Cash Flow Cycle…



* Non-Cash Item

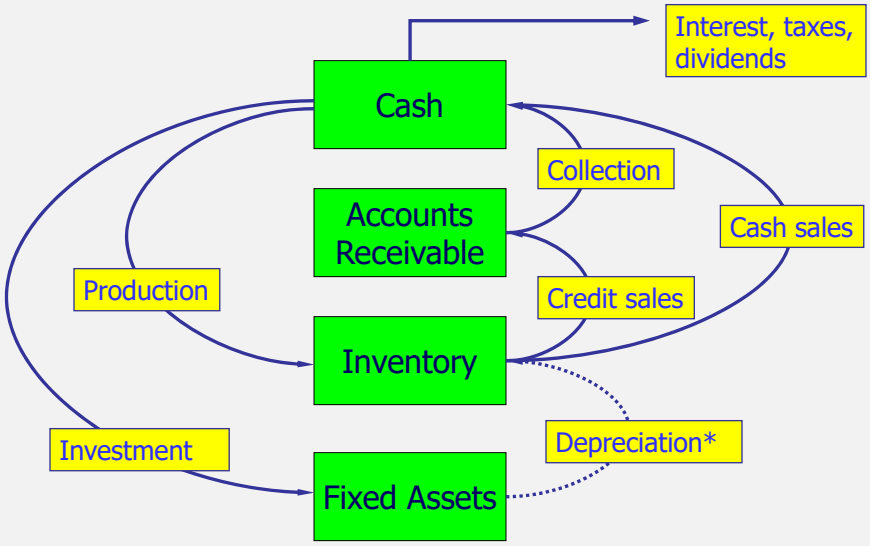
- Investment in facility: $100,000 (Year 1)
- Life of facility: 5 years
- Net Income produced by facility: $50,000/year

CASHFLOW	Year 1	Year 2	Year 3	Year 4	Year 5
Capital Investment	-$100,000				
Production Income	50,000	50,000	50,000	50,000	50,000
Net Cashflow	-$50,000	$50,000	$50,000	$50,000	$50,000

INCOME STMT	Year 1	Year 2	Year 3	Year 4	Year 5
Depreciation	-$20,000	-$20,000	-$20,000	-$20,000	-$20,000
Production Income	50,000	50,000	50,000	50,000	50,000
Net Income	$30,000	$30,000	$30,000	$30,000	$30,000

The "Matching Principle"

Cash Flow Cycle…



* Non-Cash Item

XYZ COMPANY INCOME STATEMENT

(Dollars in thousands except per-share amounts)

		Year Ended June 30,	
		2022	**2021**
1.	Net Sales	$765,000	$725,000
2.	Cost of Sales	535,000	517,000
GROSS MARGIN		230,000	208,000
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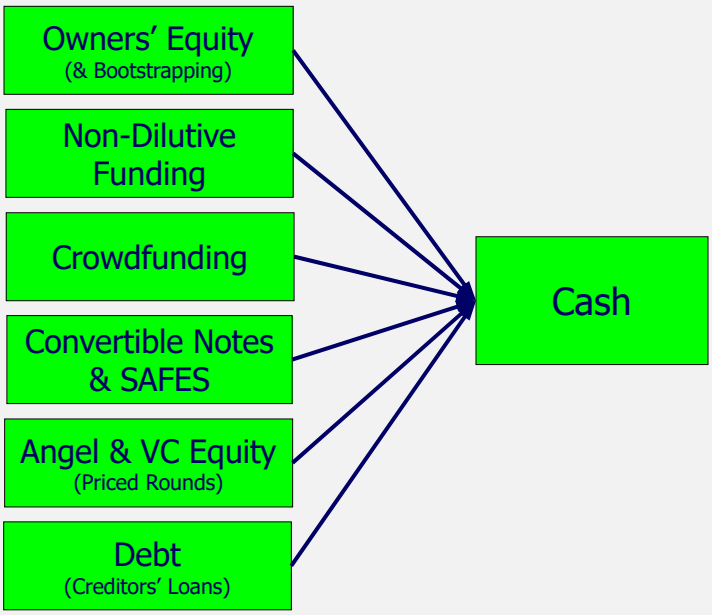
6.	Dividend and interest income	5,200	10,000
7.	Interest expense	16,250	16,750
INCOME BEFORE TAXES		94,090	66,750
8.	Income taxes	41,400	26,250
9.	Extraordinary income (loss)	0	0
NET INCOME		$52,690	$40,500

"Cash is King"

… not net income, profits or earnings.

Peter Drucker (1909-2005)

Priming the Cash Flow Cycle…



Bootstrap: Just Do It

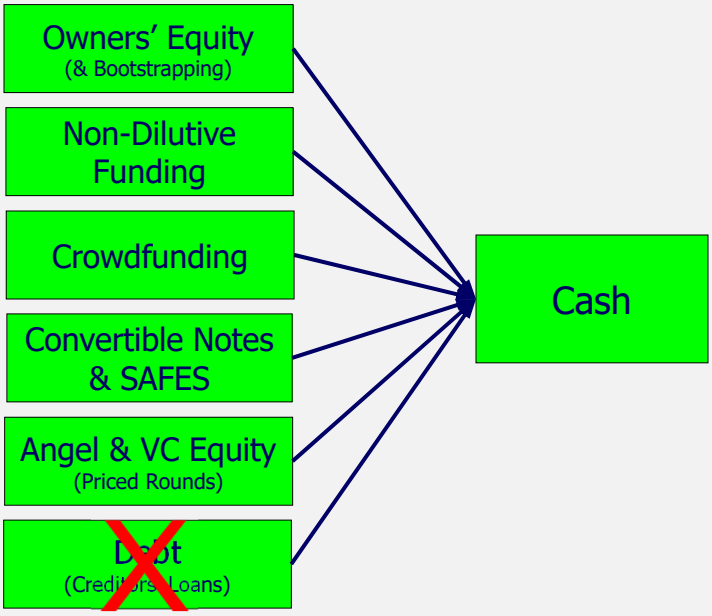
- Personal founders' funds launched **80%** of Inc. "500" companies
  - Avoids distraction from mission
  - Enhances flexibility
  - Maintains founder ownership
  - Maintains founder control
  - Doesn't "spend over" problems
  - Controlled growth

The Golden Rule:

He or she who has the gold… rules!

But, if you have no bootstrapping options…

Priming the Cash Flow Cycle…



30X more angels invested 11% of funds in 4X more deals than VC's in 2023

Angels[1]

- 422,350 active angels
- Invested $18.6B in 54,375 ventures
- Avg $340K per deal
- Earlier stage (typ.)
- 1 in 4 pitches funded

Venture Capital[2]

- 474 active V.C. funds
- $170.6B invested in 13,608 deals
- Avg $13M per deal
- Later stage (typ.)

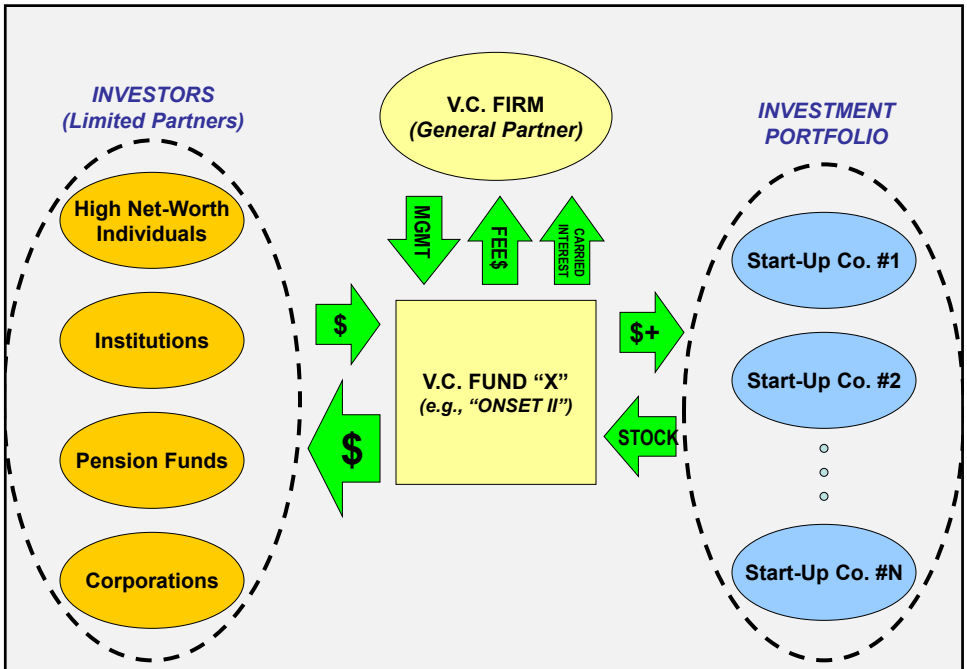
Center for Venture Research, 2023

Online Angel – Startup Meeting

AngelList

FundersClub

gust



While the choice is yours, choose investors who can (and will) provide:

- Experienced mentoring
- Sophistication, understanding
- Staying power and patience
- Future cash infusions
- Credibility
  - Access to future investors
  - Access to strategic partners

- **Be sure you can work with them… And that they will work with you.**

6 Principles of Successful Startups

ONSET VENTURES

1. Experienced mentor.
2. Flexibility with business model.
3. Defer CEO hiring until model validated.
4. Create value for next investment round.
5. There must be a *want* for the product.
6. People: missionaries not mercenaries
   - Solid understanding of technology
   - Ability to communicate articulately

Private Equity is Cyclical

- **Private equity markets = f (public stock markets)**
- **Low public markets drain money from private equity:**
  - Attractive alternative investment bargains
  - Lower investor net worth
  - Poor liquidity (IPOs, Mergers, Acquisitions)
    - Can't foresee an "exit"

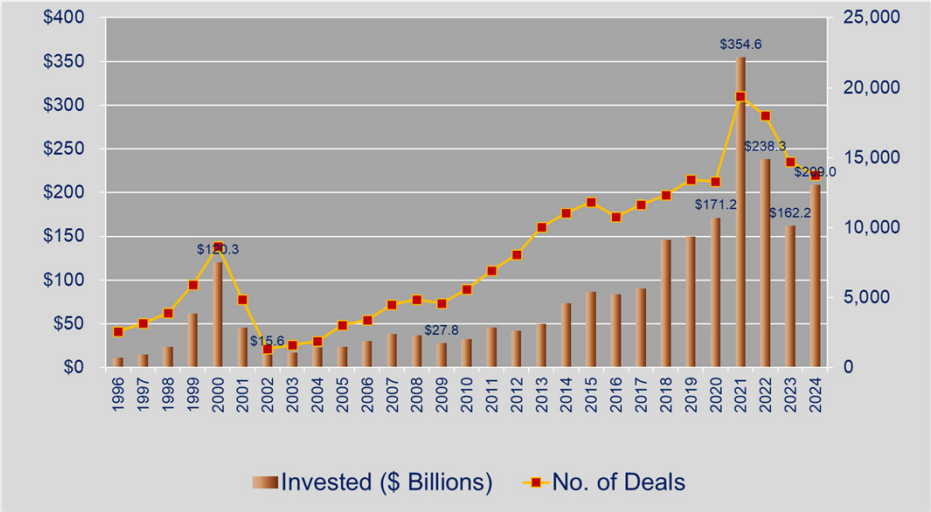
In a down (stock) market…

- **VCs fund few new deals -- preserve $ for existing portfolio**
- **VCs cannot rely on subsequent rounds:**
  - Line up future funding needs upfront
  - Milestone reviews
  - Little flexibility and patience
  - Low valuations
    - Invest less and take more

NASDAQ Performance

Yahoo Finance: Apr 28, 2025

U.S. Venture Capital



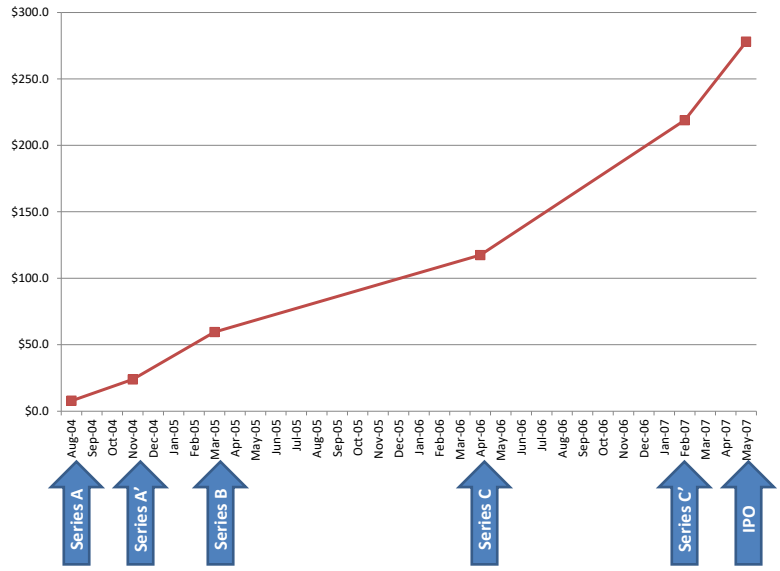
Source: PitchBook NVCA Venture Monitor, 2024

Post-Money Valuation

$$\text{Post} - \text{Money Valuation} = \frac{\text{Investment (\$)}}{\%_{\text{Acquired}}}$$

An investor offers to invest $2 million in return for 25% ownership of your company. They're saying 25% of your company is worth $2 million; therefore, they value 100% of your company at $8 million (after their money is invested).

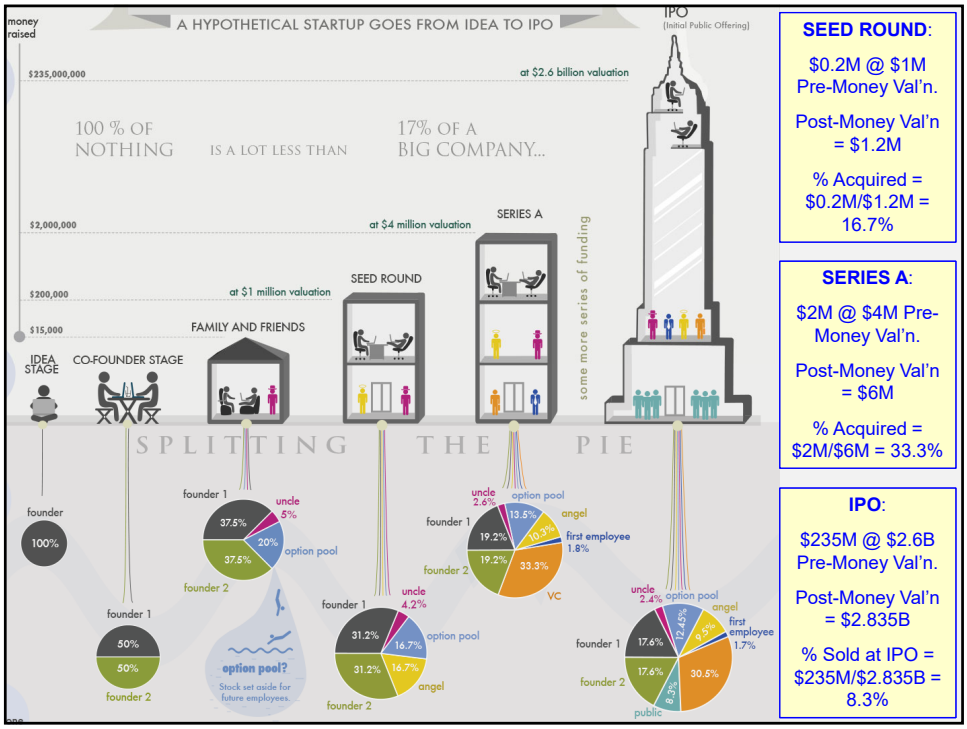
Sirtris Post-Money Valuations



Pre-Money Valuation

$$\text{Pre-Money Valuation} = (\text{Post-Money Valuation}) - (\text{Investment})$$

The investor who valued your company at $8 million after their $2 million invested, is saying your company is worth $6 million before their money is invested.



SEED ROUND:

$0.2M @ $1M Pre-Money Val'n.

Post-Money Val'n = $1.2M

% Acquired = $0.2M/$1.2M = 16.7%

SERIES A:

$2M @ $4M Pre-Money Val'n.

Post-Money Val'n = $6M

% Acquired = $2M/$6M = 33.3%

IPO:

$235M @ $2.6B Pre-Money Val'n.

Post-Money Val'n = $2.835B

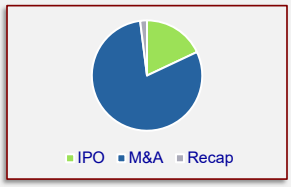
% Sold at IPO = $235M/$2.835B = 8.3%

Authorized & Issued Stock Shares

- **Shares <u>authorized</u> at time of incorporation**
  - Fixed maximum number of shares authorized for eventual issuance
  - Specified in company's Certificate of Incorporation
  - Held by company until issued
  - Par value of shares: $1/Shr typ.
- **Shares <u>issued</u> to shareholders after incorporation**
  - Drawn from authorized shares and sold to shareholders
  - Shares representing company ownership by shareholders
  - Possessed by the shareholders
  - Purchase price received by company treasury upon issuance

Investors Must Foresee an Exit Strategy

- **IPO:** Underwriters sell the company's stock in the public market.
- **Strategic Investor:** Acquiring firm buys the company's stock.
- **Recapitalization:** Company buys back its investors' stock with new debt and/or equity.



The IPO Decision

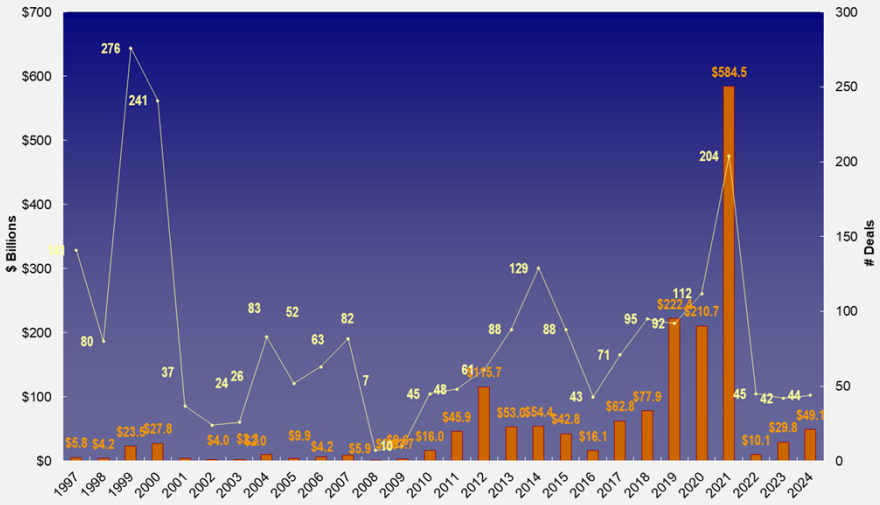
(Which may not be entirely ours to make)

- **Investor "exit" strategy…**
  … but a *transition* for us.
- **Changes dynamics of the company.**
- **Need to think long and hard:**
  - Can we go public?
  - Should we go public?
  - Do we go public?

NASDAQ Performance

Yahoo Finance: Apr 28, 2025

… and the IPO market is volatile.

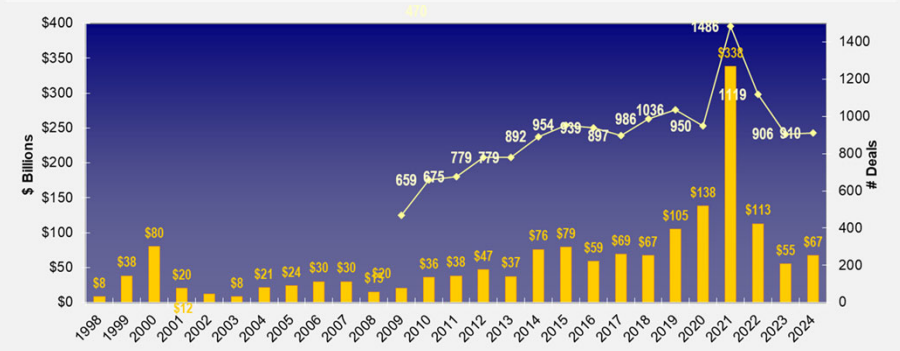


Source: Pitchbook, NVCA 2024 Yearbook

NOTE: Includes VC-backed IPOs that disclosed financial terms.

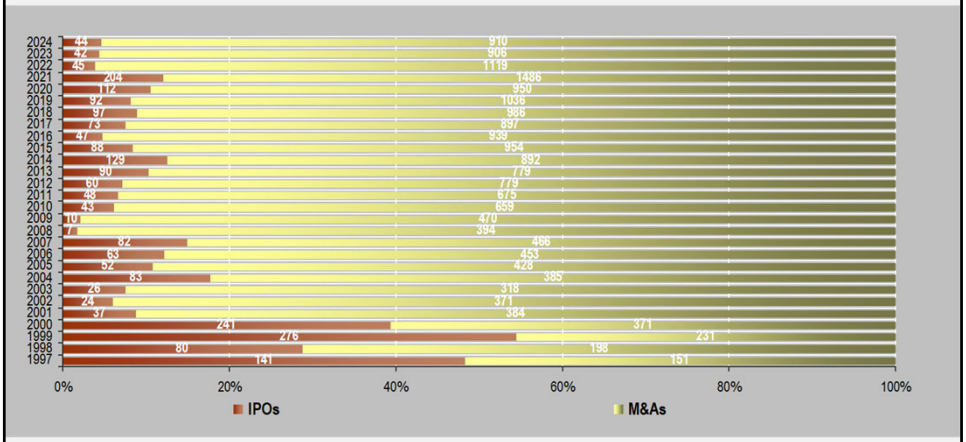
… and the M&A market is volatile

(though less volatile than the IPO market)



Source: Pitchbook, NVCA 2024 Yearbook

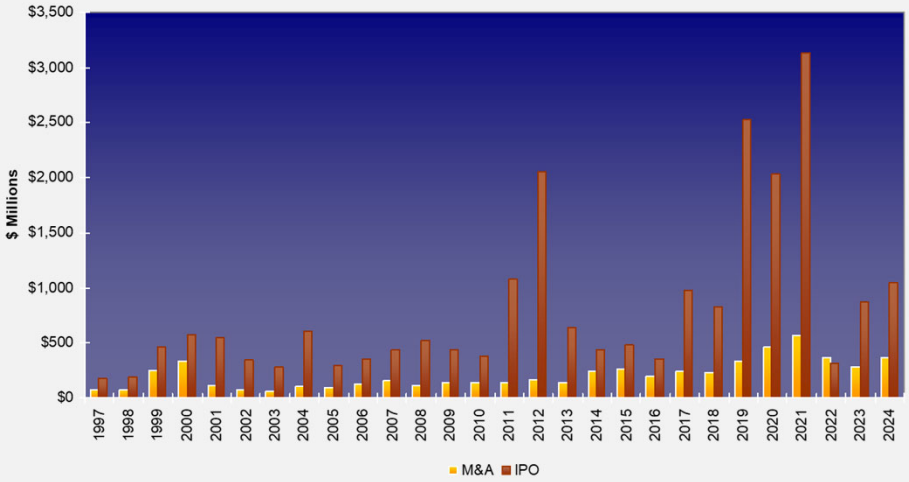
IPO's vs M&A's



Source: Pitchbook, NVCA 2024 Yearbook

IPO vs M&A Valuation

Avg. $ Millions per Deal



Source: Pitchbook 2024

GSK Acquisition Details

- April 22, 2008
- $22.50/Shr = 80% premium over $12.50/Shr market price
- $715.6 million valuation
- 25% 4-year holdback provision

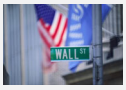
	Original Common Shares	Vested Options	Accelerated Options	Accelerated Restricted Stock	TOTAL SHARES	TOTAL VALUE ($22.50/Sh)
Christoph Westphal	355,114	173,256	409,217	207,136	**1,144,723 (3.6%)**	**$25.76MM**
David Sinclair	121,508	121,221	97,559	31,326	**371,614 (1.2%)**	**$8.36MM**

Source: SEC Schedule 14D-9 for Sirtris Pharmaceuticals

Kathy Crothall (EE '71)

Animas
Bringing new life to insulin therapy

Animas
Bringing new life to insulin therapy



May 2004: IPO @ $15.00/Shr.

Feb 2006: Acquired by J&J @ $24.50/Shr

- $525 million
- 38% premium over pre-announcement $17.75/Shr
- 6.2 X revenues in 2005

Why Go Public (Cont'd)

- **Cashing Out**
  - **Liquidity for investors**
  - **Insiders can gradually sell shares**
    - Insiders usually cannot sell much at IPO
    - Bad signal to cash out early
    - "Lock-Up Period" delays cash out
    - Risk of not being able to sell

Why Go Public (Cont'd)

- **Mergers and acquisitions**
  - Enhanced credibility
  - Shares become currency for deals
- **Higher profile image**
- **Employee compensation**
  - Can give options or equity
    - Conserves cash
    - Gives outside value to options
  - Boosts morale

IPO Concerns

- **Profit sharing**
  - Why share value with outsiders?
  - Outsiders may not understand true value
- **Public disclosure**
- **Reporting and fiduciary responsibility**
  - SEC and state securities laws
  - Filing time and expense

Going Private Is Paying Off for Dell

A year later, we're able to focus on customers and the long term, rather than activist investors.

By Michael Dell
Nov. 24, 2014 6:47 pm ET

My partners at Silver Lake Management and I successfully took Dell private a year ago in the largest corporate privatization in history. I'd say we got it right. Privatization has unleashed the passion of our team members who have the freedom to focus first on innovating for customers in a way that was not always possible when striving to meet the quarterly demands of Wall Street.

Wall Street Journal, 11/24/2014

IPO Concerns (Cont'd)

- **Time**
  - Selecting an underwriter
  - IPO preparations and road shows
  - Investor relations post-IPO
- **Expense**
  - Transaction cost
    - 7% to underwriter
    - Lawyers, accountants, printing, travel
  - Post-IPO expenses

IPO Concerns (Cont'd)

- **Loss of control**
  - Outsiders can gain control of firm
  - Hostile take-over possibilities
- **Legal liability**
  - Liable for what you SHOULD have known
  - Liable for omissions, incorrect pricing
  - Class action law suits

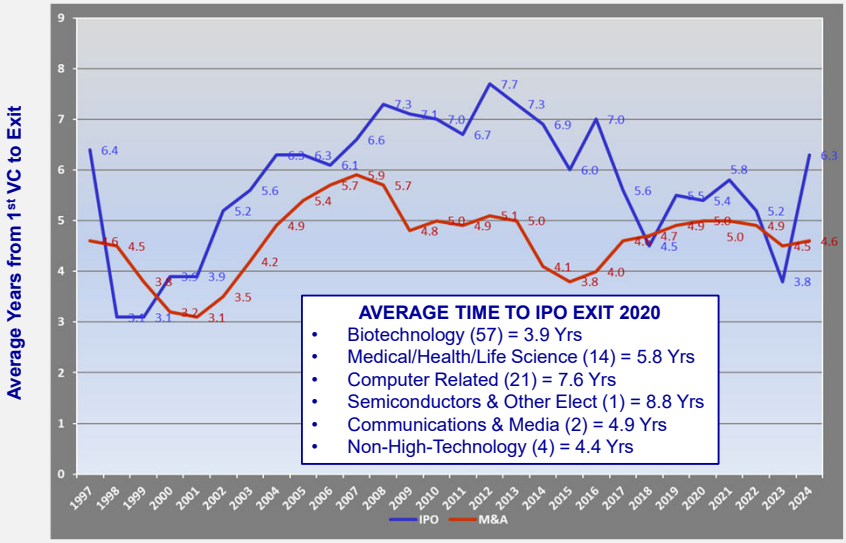
When is the Right Time?

- **Company Factors**
  - Steady growth
  - Sales revenues & earnings
  - Good story
  - Industry outlook
  - Company qualities
  - Management qualities
- **Financial Markets**
  - Window of opportunity
  - Good comparables for pricing

Many IPO firms aren't profitable yet

Source: Jay Ritter 2024 http://bear.warrington.ufl.edu/ritter/

Average Time to Exit



Source: NVCA 2024 Yearbook (Pitchbook data)

NOTE: Includes VC-backed deals that disclosed financial terms.

Underwriter(s)

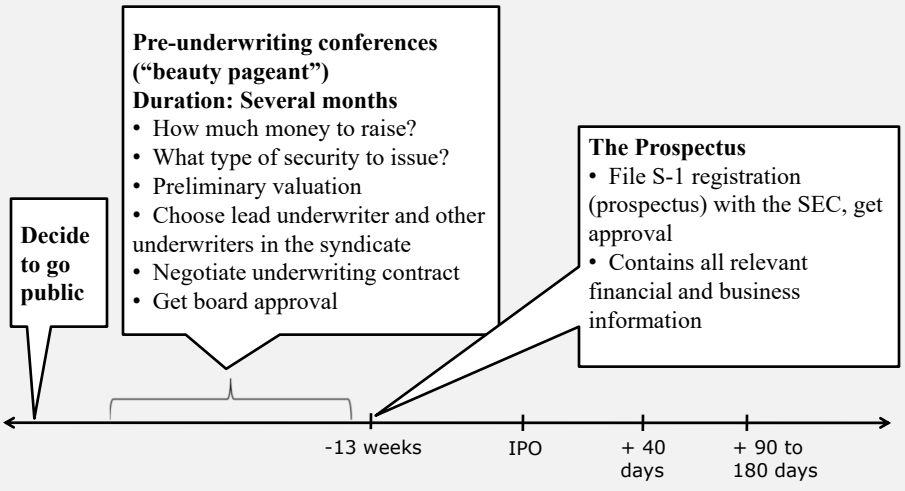
- Critical to success of the offering
- Inherent conflict of interest

Morgan Stanley

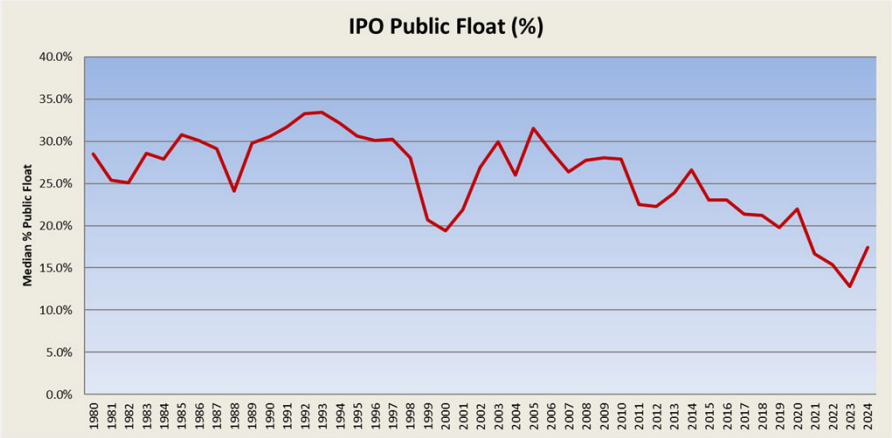
J.P.Morgan

Goldman Sachs

The IPO process



Not all Shares are Sold in the IPO

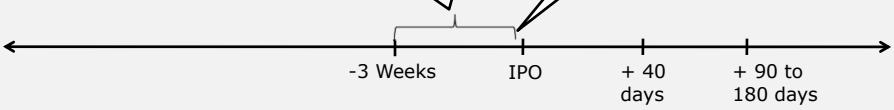


Source: Jay Ritter, IPO Statistics for 2024

The IPO process

Registration period
- Management can't talk publicly about anything not in prospectus
- Road show, or dog and pony show: Management travels around world to meet with potential investors, answer questions. Goal: Build hype so as to get a good IPO price
- Book building: Underwriters accept orders for shares from investors indicating the number of shares they want and the price they are willing to pay. Goal: Discover the right price for the IPO, and find investors to buy the IPO shares

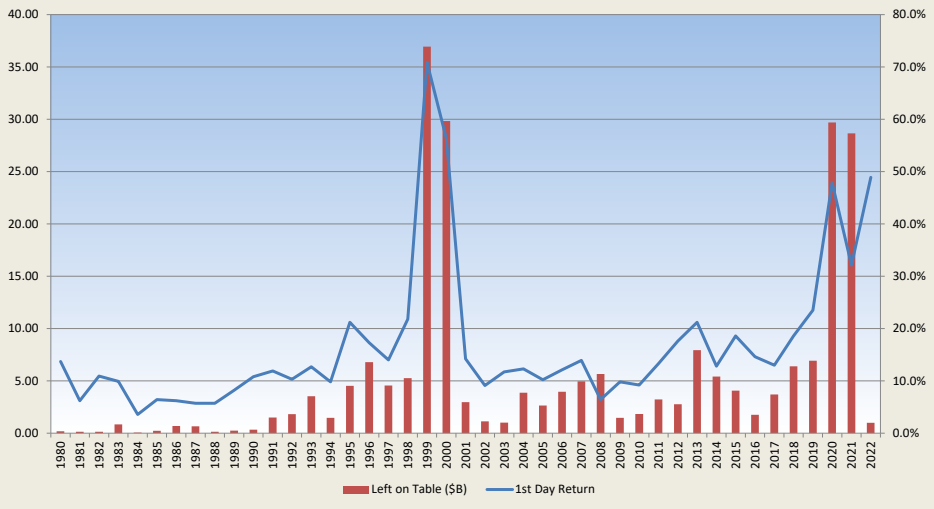
Pricing the issue
- Usually day before or morning of IPO
- Underwriters set final price for shares
- Founders / major stakeholders have option to pull IPO at last minute (rare)



IPO Pricing

1. Start with industry comparables = "book ends"
2. Price low to avoid a difficult deal
3. Road show – get orders
4. Oversubscribe X times over
5. Iterate with institutional buyers to increase price and place shares
6. Move the price up
7. Leave room for after-market bounce
8. Ultimately priced the night before it opens
   - We're a "concerned spectator"

IPO Underpricing: 1st Day "Pop"

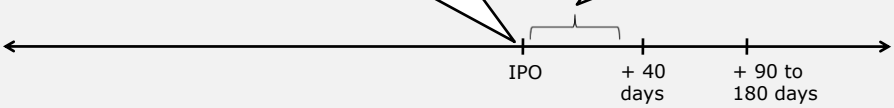


Source: Jay Ritter

The IPO process

The IPO
- Underwriters buy the shares from the company at starting price minus some discount (usually 7%)
- Underwriters sell the shares at starting price, mostly (around 80%) to institutional investors like major brokerage firms, investment banks.
- Shares then begin trading in the open market
- Usually only around 25% of company's shares are publicly traded after the IPO. Remaining shares are "locked up"

Market stabilization
- Underwriter stands ready to buy at a specified price on the market



Non-Traditional IPOs

- Google's Dutch auction with internet bidding starting at high price.
- Spotify's direct listing of non-company shares with no underwriter book sales.
- Special Purpose Acquisition Company (SPAC).

Be Wary of IPOs

- IPOs as a group perform poorly
- Conventional wisdom is to avoid IPO
- But:
  - VC (mentor) backed firms perform better
  - Larger firms perform better
  - Higher book-to-market firms perform better
  - Investors may demand it

Financial Statements

- **Income Statement**
  - Revenues – expenses = earnings
  - Over a period of time (e.g., quarterly, annually)
- **Balance Sheet**
  - Assets = liabilities + shareholders' equity
  - At a point in time (e.g., end of quarter, end of year)

XYZ COMPANY INCOME STATEMENT

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INCOME BEFORE TAXES		94,090	66,750
8.	Income taxes	41,400	26,250
9.	Extraordinary income (loss)	0	0
NET INCOME (EARNINGS)		$52,690	$40,500

XYZ COMPANY BALANCE SHEET

(Dollars in thousands)

		June 30, 2022	June 30, 2021
ASSETS			
12.	Cash and cash equivalents	$19,500	$15,000
13.	Marketable securities	46,300	32,000
14.	Accounts receivable	162,000	150,000
15.	Less: allowances for doubtful accounts	(6,000)	(5,000)
16.	Inventories	180,000	185,000
17.	Prepaid expenses and other current assets	4,000	3,000
	TOTAL CURRENT ASSETS	405,800	380,000
18.	Property, plant and equipment	385,000	346,600
19.	Less: accumulated depreciation	(125,000)	(97,000)
	NET PROPERTY, PLANT & EQUIPMENT	260,000	249,600
20.	Intangible assets (goodwill, patents, etc.)	2,400	2,400
21.	Less: accumulated amortization	(450)	(390)
	NET INTANGIBLES	1,950	2,010
22.	Other assets	0	0
TOTAL ASSETS		$667,750	$631,610
LIABILITIES AND SHAREHOLDERS' EQUITY			
23.	Accounts payable	$72,000	$69,000
24.	Notes payable	51,000	61,000
25.	Other accrued expenses	30,000	36,000
26.	Income taxes payable	17,000	15,000
27.	Current portion of long-term debt	6,000	
	TOTAL CURRENT LIABILITIES	176,000	181,000
28.	Long-term debt	130,000	136,000
29.	Deferred income taxes	16,000	9,000
TOTAL LIABILITIES		$322,000	$326,000
30.	Preferred stock (7.5% cumulative): $100 par value	6,000	6,000
31.	Common stock: $5 par value (20,000,000 shares authorized)	75,000	72,500
	15,000,000 shares issued and outstanding 2007		
	14,500,000 shares issued and outstanding 2006		
32.	Additional paid-in capital	20,000	13,500
33.	Retained earnings	244,750	213,610
TOTAL SHAREHOLDERS' EQUITY		$345,750	$305,610
TOTAL LIABILITIES AND SHAREHOLDERS' EQUITY		$667,750	$631,610

=

Assets

- All property owned by the company including uncollected amounts (“Receivables”) due from customers.
- “**Current Assets**” include cash and assets that will be turned into cash within the next 12 months.

Liabilities

- All debts and amounts owed by the company to others.
- "**Current Liabilities**" include obligations of the company due and payable within the next 12 months.

Key Financial Ratios

XYZ COMPANY INCOME STATEMENT
(Dollars in thousands except per-share amounts)

		Year Ended June 30,	
		2007	2006
1.	Net Sales	$765,000	$725,000
2.	Cost of Sales	535,000	517,000
GROSS MARGIN		230,000	208,000
3.	Selling expenses	76,500	84,000
4.	General and administrative expenses	20,300	25,500
5.	Depreciation and amortization	28,060	25,000
OPERATING INCOME		105,140	73,500
6.	Dividend and interest income	5,200	10,000
7.	Interest expense	16,250	16,750
INCOME BEFORE TAXES		94,090	66,750
8.	Income taxes	41,400	26,250
9.	Extraordinary income (loss)	0	0
NET INCOME		$52,690	$40,500

XYZ COMPANY BALANCE SHEET
(Dollars in thousands)

		June 30,	
		2007	2006
ASSETS			
12.	Cash and cash equivalents	$19,500	$15,000
13.	Marketable securities	46,300	32,000
14.	Accounts receivable	162,000	150,000
15.	Less: allowances for doubtful accounts	(6,000)	(5,000)
16.	Inventories	180,000	185,000
17.	Prepaid expenses and other current assets	4,000	3,000
TOTAL CURRENT ASSETS		405,800	380,000
18.	Property, plant and equipment	385,000	346,600
19.	Less: accumulated depreciation	(125,000)	(97,000)
NET PROPERTY, PLANT & EQUIPMENT		260,000	249,600
20.	Intangible assets (goodwill, patents, etc.)	2,400	2,400
21.	Less: accumulated amortization	(450)	(390)
NET INTANGIBLES		1,950	2,010
22.	Other assets	0	0
TOTAL ASSETS		$667,750	$631,610
LIABILITIES AND SHAREHOLDERS' EQUITY			
23.	Accounts payable	$72,000	$69,000
24.	Notes payable	51,000	61,000
25.	Other accrued expenses	30,000	36,000
26.	Income taxes payable	17,000	15,000
27.	Current portion of long-term debt	6,000	
TOTAL CURRENT LIABILITIES		176,000	181,000
28.	Long-term debt	130,000	136,000
29.	Deferred income taxes	16,000	9,000
TOTAL LIABILITIES		$322,000	$326,000
30.	Preferred stock (7.5% cumulative): $100 par value	6,000	6,000
31.	Common stock: $5 par value (20,000,000 shares authorized) 15,000,000 shares issued and outstanding 2007 14,500,000 shares issued and outstanding 2006	75,000	72,500
32.	Additional paid-in capital	20,000	13,500
33.	Retained earnings	244,750	213,610
TOTAL SHAREHOLDERS' EQUITY		$345,750	$305,610
TOTAL LIABILITIES AND SHAREHOLDERS' EQUITY		$667,750	$631,610

- **INVENTORY TURNOVER:**

$$\frac{\text{COGS}}{\text{Inventory}} \quad \text{or} \quad \frac{365}{\text{COGS/Inventory}}$$

XYZ Company 2007

$$\frac{535}{180} = 3X \text{ or } 123 \text{ Days}$$

vs 131 Days in 2006

- **COLLECTION PERIOD:**

$$\frac{\text{Accts Receivable}}{\text{Sales/365}}$$

$$\frac{162}{765/365} = 77 \text{ Days}$$

vs 76 Days in 2006

- **DAYS OF PAYABLES:**

$$\frac{\text{Accts Payable}}{\text{COGS/365}}$$

$$\frac{72}{535/365} = 49 \text{ Days}$$

vs 49 Days in 2006

- **CURRENT RATIO:**

$$\frac{\text{Current Assets}}{\text{Current Liabilities}}$$

XYZ Company 2007

$$\frac{406}{176} = 2.3\ X$$

vs 2.1X in 2006

- **QUICK RATIO ("ACID TEST"):**

$$\frac{(\text{Current Assets}) - (\text{Inventory})}{\text{Current Liabilities}}$$

$$\frac{406 - 180}{176} = 1.3\ X$$

vs 1.1X in 2006

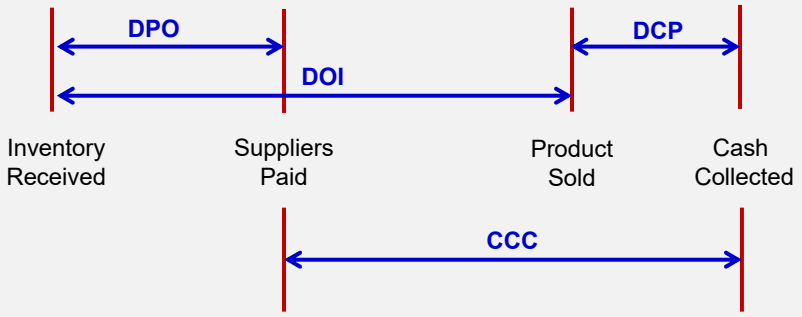
- **RETURN ON EQUITY (ROE):**

$$\frac{\text{Earnings}}{\text{Shareholders' Equity}}$$

$$\frac{52.7}{346} = 15.2\%\ p.a.$$

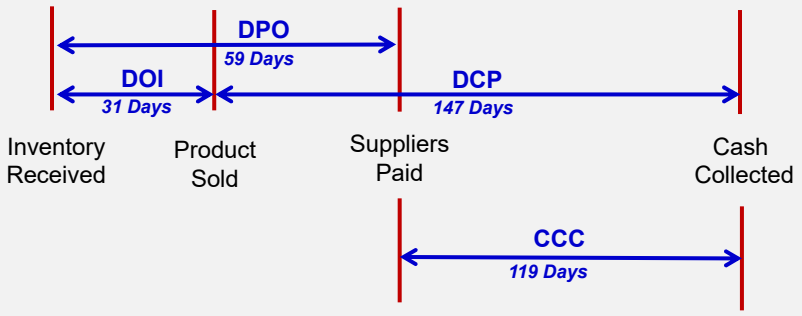
vs 13.3% in 2006

Cash Conversion Cycle



- DPO: Days Payables Outstanding = Accts Payable / (COGS/365)
- DOI: Days of Inventory = Inventory / (COGS/365)
- DCP: Days Collection Period = Accts Receivable / (Sales/365)
- **CCC = DOI + DCP - DPO**

Cash Conversion Cycle



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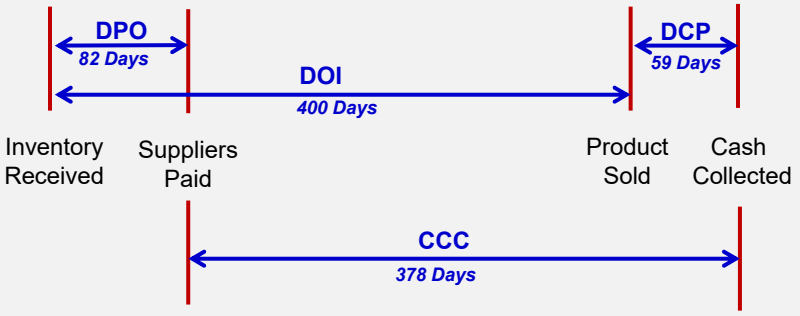
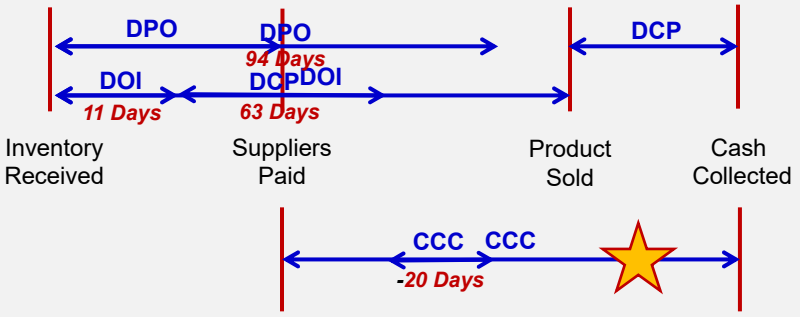
Cash Conversion Cycle



- DPO: Days Payables Outstanding
- DOI: Days of Inventory
- DCP: Days Collection Period
- **CCC = DOI + DCP - DPO**

Cash Conversion Cycle

Dell Computer 2013



- DPO: Days Payables Outstanding = Accts Payable / (COGS/365)
- DOI: Days of Inventory = Inventory / (COGS/365)
- DCP: Days Collection Period = Accts Receivable / (Sales/365)
- **CCC = DOI + DCP - DPO**

Pricing Strategy

- Objectives
  - Increase sales, build traffic, establish market position
  - Increase profits
    - Short-term
    - Long-term
  - Discourage entrants, injure marginal competitors
  - Stabilize market prices
  - Recover product development costs
  - Enhance image
  - Enhance reputation of fairness to channels / customers
- Strategies
  - Price high: skimming
  - Price low: penetration
  - Meet the competition: going rate
- Variables
  - List price
  - Credit terms
  - Discounts
  - Allowances
  - Flexibility
- = f (time)

"Managers would calculate how much it cost to make and deliver each product and add a flat percentage on top."

"They had to start determining prices by what a customer is willing to pay rather than what a product costs to make."

THE WALL STREET JOURNAL.

CHANGING THE FORMULA

Seeking Perfect Prices, CEO Tears Up the Rules

Parker's Washkewicz Weighs Market Power Of 800,000 Parts

By Timothy Aeppel

CLEVELAND—In early 2001, shortly after Donald Washkewicz took over as chief executive of Parker Hannifin Corp., he came to an unnerving conclusion. The big industrial-parts maker's pricing scheme was crazy.

For as long as anyone at the 89-year-old company could recall, Parker used the same simple formula to determine prices of its 800,000 parts from heat-resistant seals for jet engines to steel valves that hoist buckets on cherry pickers. Company managers would calculate how much it cost to make and deliver each product and add a flat percentage on top, usually aiming for about 35%. Many managers liked the method because it was straightforward and gave them broad authority to negotiate deals.

But Mr. Washkewicz thought that Parker, which had revenues of $9.4 billion last year, had stuck itself in a profit-margin rut. No matter how much a product improved, the company often ended up charging the same premium it would for a more standard item. And if the company found a way to make a product less expensively, it ultimately cut the product's price as well.

"I was actually losing sleep," recalls Mr. Washkewicz, a 56-year-old Cleveland native who started with the company when he was 22 and rose through its ranks as an engineer.

While touring the company's 225 facilities in 2001, Mr. Washkewicz had an epiphany: Parker had to stop thinking like a widget maker and start thinking like a retailer, determining prices by what a customer is willing to pay rather than what a product costs to make. Such "strategic" pricing schemes are used by many different industries. Airlines know they can get away charging more for a seat to Florida in January than in August. Sports teams raise ticket prices if they're playing a well-known opponent. Why shouldn't Parker do the same, Mr. Washkewicz reasoned.

Donald Washkewicz

Today, the company says its new pricing approach boosted operating income by $200 million since 2002. That helped Parker's net income soar to $673 million last year from $130 million in 2002. Now, the company's return on invested capital has risen from 7% in 2002 to 21% in 2006, putting it on the verge of moving into the top 25% of Mr. Washkewicz's list comparing Parker with "peer" industrial companies.

From the end of 2001 to present,

Please turn to page A16

Wall Street Journal, Feb 27, 2007

"The company's new strategic pricing approach boosted operating income by $200 million since 2002."

"Return on invested capital has risen from 7% in 2002 to 21% in 2006."

"The company's shares have risen nearly 88% compared to a 25% gain in the S&P 500."

Top Down, not Bottom Up!

Value-based not Cost-based

- **Price without regard to cost.**
- **Price = f (positioning):**
  - Product differentiation
  - Target market segment
- **8 Steps to Better Pricing**
- **Test the price point:**
  - Concept testing.
  - Conjoint analysis.

Venture Capital Method

VC Problem Set 202510

Study the Note on Venture Capital Method in Canvas Files

VC Problem Set 2510

- **2024 Company Launch**
  - 1,000,000 Founders Shares
    - Ken: 350,000 shrs for $30,000
    - Two Co-Founders: 150,000 shrs for $20,000
- **3 Rounds of Financing (per Business Plan)**
  - 2024 seed: $400,000 post-money SAFE (Cathy)
    - 20% conversion discount from Series A pricing
    - $4,000,000 post-money valuation cap
    - $1,000,000 minimum Series A to trigger conversion
  - 2025 Series A: $1,500,000 (TVC @ 50% p.a. ROI)
  - 2027 Series B: $7,000,000 (TVC @ 40% p.a. ROI)
  - 2028 Series C: $8,500,000 (TVC @ 30% p.a. ROI)
- **2030 Terminal Valuation for Strategic Acquisition**
  - $6,500,000 annual revenues (projected in Business Plan)
  - $750,000 annual earnings (projected in Business Plan)
  - P/R = 14:1

- **Questions 1a & 1b**
  - % acquired by TVC for Series A, B & C
  - # shares for Series A, B & C
  - Price/share for Series A,B & C
  - Pre- and post-money valuations for Series A, B & C

Company Valuation

"Terminal Value at Exit in Year T: V_T"

- Price/Earnings Multiple (P/E):

  $$V_T = P/E \times E_T$$

  where E_T = projected net earnings in year T.

OR

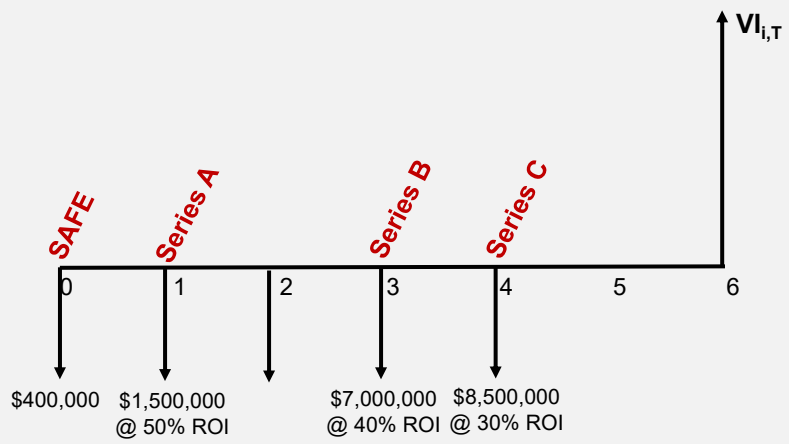
- Price/Revenues Multiple (P/R):

  $$V_T = P/R \times R_T$$

  where R_T = projected sales revenues in year T.

Multiple Investment Rounds

$\boxed{VI_{i,T} = I_i\,(1 + ROI_i)^{(T-i)}}$ **OR** $VI_{i,T} = I_i\,X_i$



Terminal Ownership

- % of company investor must own at the time (T) of the exit to achieve target ROI:

$$(\%\ Ownership)_i = \frac{VI_{i,T}}{V_T}$$

Retention

$$\text{Retention } \% = 100\% - \sum \left(\%_{\text{Ownership}} \text{ of Later Rounds} \right)$$

% Acquired at Time of Investment

- % of company investor must acquire at the time of investment to achieve target ROI:

$$\%_{\text{Acquired}} = \frac{\left(\%_{\text{Ownership (Terminal)}} \right)}{\left(\text{Retention } \% \div 100\% \right)}$$

Post-Money Valuation

$$\text{Post-Money Valuation} = \frac{\text{Investment (\$)}}{\%_{\text{Acquired}}}$$

Pre-Money Valuation

$$\text{Pre-Money Valuation} = (\text{Post-Money Valuation}) - (\text{Investment})$$

Post-Money SAFE

- Principal Amount of SAFE Investment
- Minimum Investment of Next Round Triggering Conversion
- Price/Shr Discount When Converting (None or 20% typ.)
- Post-Money Valuation Cap on Conversion
- Right to Purchase Next Round of Financing
- If Acquisition Prior to Next Round, Premium Over Principal
- Board Seat(s) Not Uncommon

Number & Price of Converted Shares

- Post-Money Valuation of the SAFE =
  (Pre-Money Valuation of Series A) x (1-Discount)

- $(\%\ \text{Acquired})_{SAFE} = \dfrac{(\text{Principal Amount of SAFE})}{(\text{Post-Money Valuation of SAFE})}$

  where the Post-Money Valuation of the SAFE in the denominator is the lesser of the Post-Money Valuation calculated above or the Post-Money Valuation Cap.

- $\#\ \text{Shares Acquired}\ (S_{SAFE}) = \dfrac{\%_{\text{Acquired}}}{1-\%_{\text{Acquired}}} \times (\#\ \text{Old Shares})$

- $\text{Price per share}\ (P_{SAFE}) = = \dfrac{(\text{Principal Amount of SAFE})}{(S_{SAFE})}$

Shares Purchased

New Shares Issued for Series A & B Investments

$$\%_{\text{Acquired}} = \frac{\#\ \text{Shares Purchased}}{\text{Total}\ \#\ \text{Shares Issued}}$$

$$\%_{\text{Acquired}} = \frac{\#\ \text{Shares Purchased}}{\#\ \text{Old Shares}^{*} + \#\ \text{Shares Purchased}}$$

$$\#\ \text{Shares Purchased} = \frac{\%_{\text{Acquired}}}{\left(1-\%_{\text{Acquired}}\right)} \times \left(\#\ \text{Old Shares}\right)$$

* # Old Shares = Number of shares issued pre-investment

Price per Share

$$\text{Price per Share} = \frac{\text{Investment (\$)}}{\text{\#Shares Purchased}}$$

Terminal Price per Share

$$Price\ per\ Share = \frac{Company\ Value, V_T}{Total\ \#Shares\ Issued}$$

Series A & B Values at Exit

Distribution = $(\text{Price/Shr})_{\text{terminal}}$ x (# Shares)

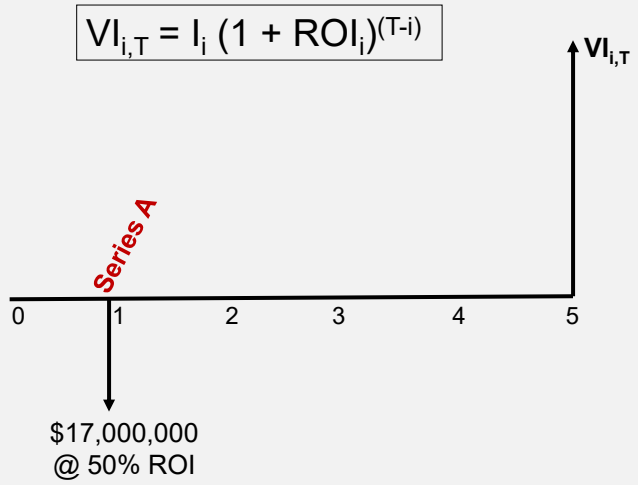
Return on Investment

$$\text{ROI} = \left(\frac{\text{Price per share}_{\text{sold}}}{\text{Price per share}_{\text{purchased}}} \right)^{1/N} - 1$$

- **Question 2**
  - At time of exit, what will Ken's and co-founders' shares be worth and what will be their respective compounded rates of return?
  - Explain difference between Ken's and co-founder's rates of return.
  - At time of exit, what will Cathy's SAFE shares be worth and what will be her compounded rate of return?

- **Question 3**
  - Raise all money in a single priced round in 2025 @ 50% target ROI

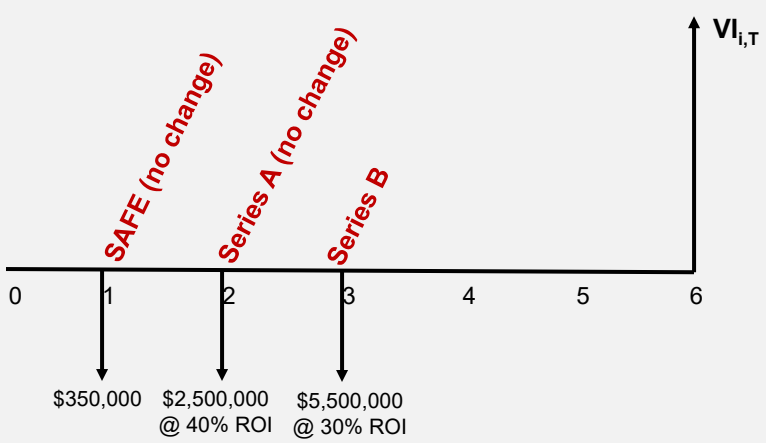
$$VI_{i,T} = I_i\,(1 + ROI_i)^{(T-i)}$$



- **Question 4**
  - **Include Option Pool equal to 10% of the company at the time of exit.**

$$\text{Retention } \% = 100\% - \sum \left(\%_{\text{Ownership}} \text{ of Later Rounds} \right)$$

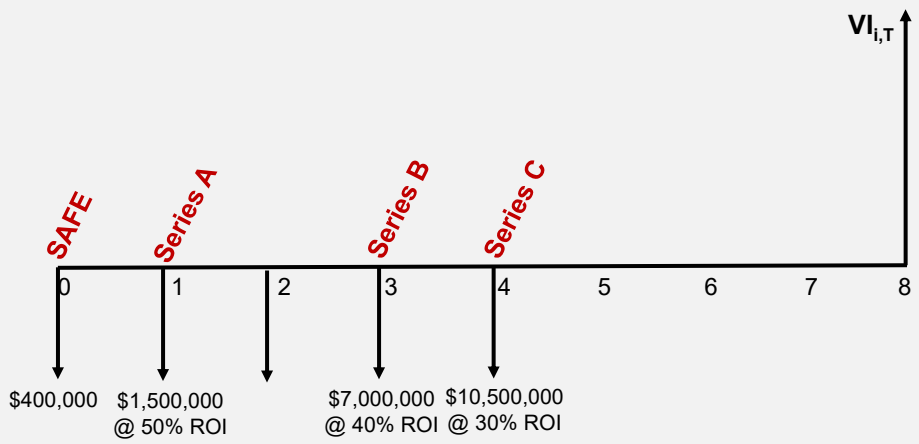
$$VI_{\text{Series B}} = I_B \, (1 + ROI_B)^{(T-3)}$$



Question 5

- After Series A but before Series B, learned that time of exit delayed two years and $2,000,000 more needed in Series C.

$$VI_{i,T} = I_i\,(1 + ROI_i)^{(T-i)}$$



Question 6

- Series B & C are participating preferred shares.

Distribution =

$$\text{(Investment)} + \left\{\left[(\text{Value of Company}) - (\text{Investment})\right] \text{ x } \left[\frac{\text{Total Shares Owned}}{\text{Total Shares Issued}}\right]\right\}$$

Evaluating Venture Capital Term Sheets

EAS5450

Penn Engineering

WALNUT Venture Associates

Investment Process

Pre-Screening

- Geography
- Industry
- Potential For High Returns
- Initial Assessment of Business Idea/Model
- Initial Assessment of Entrepreneur
- Initial Assessment of Management Team

Tuesday Presentation

- 30 Minute Presentation to Walnut Group
  - 15 Min. Presentation
  - 15 Min. Q&A
- Problem Solved
- Market Size/Growth
- Solution
- Initial Markets and/or Customers
- Team Track Record
- Business Model
- Capital Needs and Milestones
  - Now
  - Next Rounds

Due Diligence

- Follow-up Meeting: Meet with Key Management at Company Office
- Thorough Due Diligence
  - Personal References
  - Customer References
  - Competitive Assessment
  - Financial Review
- Negotiate Term Sheet

Investment and Support

- Board Seat
- Refine your business plan
- Help Recruit Key Management
- Introduce Company to Sources of Follow-on Financing
- Guidance on Financial Strategy
- Make key contacts for strategic partnerships
- Configure the company for an IPO or being acquired

In the Tuesday Meeting, the Company Should Concisely Cover, in 15 Minutes, the Key Areas that will Drive the Company's Success; Allow 15 Minutes for Q&A

Introduction and Concept

- Elevator Pitch- Briefly tell us what important problem you solve and how you will make money
- Concept- Describe the product or service you will offer and how it solves an important problem

Markets and Customers

- Demonstrate there is a rapidly growing market and briefly describe one or two driving forces
- Customers: Who will buy this and what is it worth to them?
- How will you reach/sell to the customers?

Management Team

- Tell us why your team can execute and win
  - Relevant industry experience
  - Customer knowledge/contacts
  - Technology expertise
  - Company/organization building skills
  - Other experience or skills you will need to bring in

Competition

- What are the alternative ways of solving the problem?
  - Who are the competitors?
- How does your offering differ from the others?
- Tell us why and how you can win

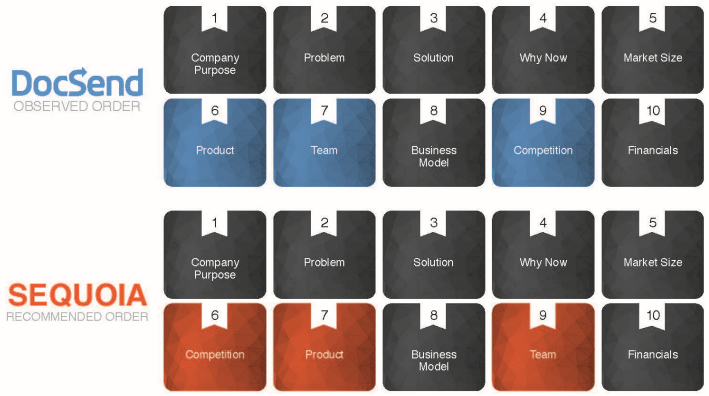
Finances and Capital Needs

- Show high level revenues and costs, and include the major driving assumptions of your business model
- Tell us how much total capital you will need to achieve cash flow breakeven and profitability; and when you will need these financing rounds over time
- How much capital do you need for this financing and what do you feel is the valuation of you company
- How will investors exit?
  - IPO-why is this a good company for an IPO
  - Being acquired- who would buy and why

Other

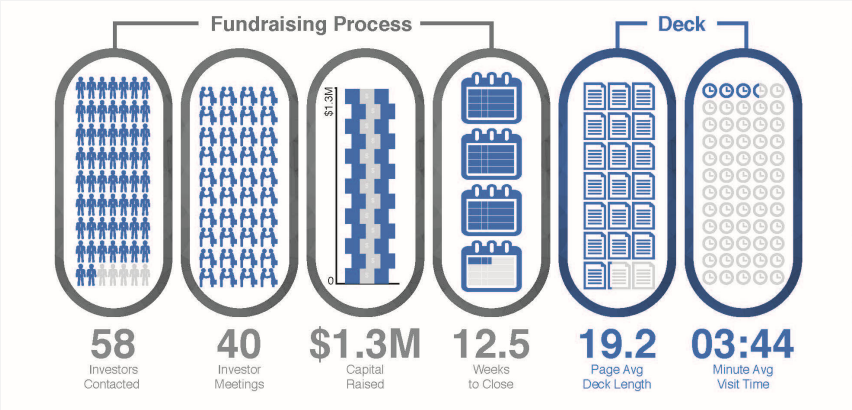
- Are there any crucial alliances required to execute the business model such as distribution or technology?

Designing the 15-minute Pitch



The ordering of the sections was fairly standard, except for the team page. This page appeared either at the beginning or the end, but never in the middle. Within the 200 companies we studied, more companies placed the page at the end and not the beginning. This practice is in line with advice from Reid Hoffman, who advises companies to lead with their investment thesis, not a team page.

An Average Series Seed Raise



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Term Sheet

- **Provided by an interested investor to the entrepreneur.**
- **Not a legally binding contract.**
  - Outlines the terms for an eventual legally binding definitive agreement.
  - Contingent upon due diligence.
- **An agreement to negotiate in good faith.**
  - Often provides for a “standstill” period of exclusivity.

3 "Hot" Provisions

- **Valuation (Pricing)**
- **Governance:**
  The Golden Rule
  - Voting rights
  - Board seats
  - Committees
  - Protective provisions
- **Capital Structure**
  - Type of security
  - Liquidation preferences
  - Conversion rights
  - Dividend preferences
  - Redemption rights
  - Anti-dilution provisions

Anti-Dilution: Full Ratchet

1. Founders: 2,000,000 Shrs
2. Investor #1: $4,000,000 = 1,000,000 Shrs @ $4/Shr
3. Investor #2 down-round: $2/Shr
4. Investor #1 re-priced @ $2/Shr:

$$\frac{\$4{,}000{,}000}{\$2/\text{Shr}} = 2{,}000{,}000 \text{ Shrs}$$

➡ **+ 1,000,000 Shrs for Investor #1**

Anti-Dilution: Weighted Ratchet

1. Founders: 2,000,000 Shrs
2. Investor #1: $4,000,000 = 1,000,000 Shrs @ $4/Shr
3. Investor #2: $1,200,000 = 600,000 Shrs @ $2/Shr
4. Investor #1 re-priced:

$$P_{1,New} = P_{1,Old} \times \left[\frac{(\text{Old Shrs}) + \left(\frac{\text{Investment}_2}{P_{1,Old}}\right)}{(\text{Old Shrs}) + \left(\frac{\text{Investment}_2}{P_2}\right)}\right] = P_{1,Old} \times \left[\frac{\text{Total shares after \#2 if priced at } P_{1,Old}}{\text{Total shares after \#2 if priced at } P_2}\right]$$

$$P_{1,New} = \$4 \times \left[\frac{3{,}000{,}000 + \left(\frac{\$1{,}200{,}000}{\$4}\right)}{3{,}000{,}000 + \left(\frac{\$1{,}200{,}000}{\$2}\right)}\right] = \$3.67$$

$$\boxed{\frac{\$4{,}000{,}000}{\$3.67\,/\,\text{Shr}} = 1{,}090{,}000 \text{ Shrs}}$$

➡ **+ 90,000 Shrs for Investor #1**

Provisions Affecting Founder's Position

- **Vesting**
- **Performance/Forfeiture**
- **Employment Contracts**
  - Non-compete
  - Severance/termination
- **Shareholder Agreements**
  - New issue restrictions
  - Sale restrictions

Information Provisions

- **Disclosure**
  - Representations & warranties
- **Information Rights**
  - Financial records & statements
  - Business plans
  - Inspection & visitation rights

Cashing Out Provisions

- **Co-Sale**
- **Right of First Refusal**
  - Future investment rounds
  - Other shareholder's sale of stock
- **Drag Along Rights**
- **Registration Rights**
  - "Piggy-Back"
  - Demand rights
  - Frequency, timing and expense
- **Liquidation/Acquisition/Merger**

Other Provisions

- **Key Man Insurance**
  - Company pays
  - Investors benefit
- **Conditions Precedent**

Key Man Insurance Example

"Life Insurance Policy" means the policy of life insurance issued to the Company by The Manufacturers Life Insurance Company of America covering the life of Thomas A.V. Cassel in the face amount of $5,000,000.

Section 5.8 LIFE INSURANCE POLICY. The Company shall pay all premiums when due on the Life Insurance Policy and promptly notify the Bank in writing of the death of any individual covered by a Life Insurance Policy; in the event of the death of any such individual, regardless of whether an Event of Default shall have occurred and be continuing, the Company agrees that the Bank may, upon three (3) Business Days' prior written notice to the Company, apply the proceeds of such Life Insurance Policy as a prepayment of the Loans and any and all other amounts payable hereunder in such order of priority as the Bank may determine in its discretion.

Evaluating Venture Capital Term Sheets

Universal MobileApps, Inc.

John Stevens & Edward Lopez, Co-Founders

TOP GUN

- Series A Preferred
- $4,000,000
- 100% Top Gun
- $9,000,000 post-money valuation
- 25% employee option pool (post-closing)
- Price: $1.00/share
- Weighted ratchet anti-dilution
- Dividend: 8% p.a.
- 1 of 3 board seats
- Founder vesting: ?
- Pay to play: no
- Fees for diligence: up to $30,000
- Key person insurance: No

RED BARON

- Series A Preferred
- $6,000,000
- Split round: 50% Red Baron, 50% other (?)
- $12,000,000 post-money valuation
- 15% employee option pool (post-closing)
- Price: $1.00/share
- Full ratchet anti-dilution
- Dividend: ?
- 2 of 3 board seats
- Founder vesting: 4 years
- Pay to play: yes
- Fees for diligence: up to $40,000
- Key person insurance: Yes

Evaluating Term Sheets

	Top Gun	Red Baron
Valuation	$5M Pre-Money	$6M Pre-Money
Governance	1 of 3 Board Seats	2 of 3 Board Seats
Security Type	Preferred	Participating
Employee Stock Options	25%	15%
Investors' Equity Share	48.5%	53.3%
Founders' Equity Share	26.5%	31.7%
Split Round?	Entire Round	Half of Round
Anti-Dilution	Weighted Average	Full Ratchet
Founder Vesting	---	4 Years
Pay To Play	No	Yes
Fees	Up to $30,000	Up to $40,000
Key Person Insurance	No	Yes
What Comes With the Money	Relevant Experience	Young Firm

Exam Format - All Sections

Tuesday, May 6th, 12-2PM

- **Section 1: Short Answer (80%)**
  - Answer 16 of 20 questions requiring brief (1-4) sentence response and/or calculations
  - Bullets OK if question asks for list
- **Section 2: VC Evaluation (20%)**
  - Choose from two term sheet offers
  - Answer all 8 of 8 questions
  - Quantitative analysis
  - Qualitative analysis